FastAPI

Running FastAPI with Uvicorn, Gunicorn

Quick Recap

- What is FastAPI
- FastAPI vs Flask vs Django
- ASGI vs WSGI
- First GET API
- Running app

Quick Quiz

- 1. Which command is used to run FastAPI?
- 2. Which decorator is used for GET API?
- 3. FastAPI is based on which standard? (ASGI/WSGI)
- 4. Which library is used to run FastAPI server?
- 5. What URL is used to access FastAPI by default?

What is Uvicorn?

- ASGI server
- Runs FastAPI applications
- High performance
- Production ready

Why Uvicorn?

- Async support
- Fast execution
- Handles concurrency
- Recommended for FastAPI

Running FastAPI with Uvicorn

- uvicorn main:app

Uvicorn with Reload

- uvicorn main:app --reload
- Auto reload on code change

Uvicorn Parameters

- --host
- --port
- --reload
- --workers

Host & Port Example

- uvicorn main:app --host 0.0.0.0 --port 8000

Workers Parameter

- Multiple workers for performance
- Example:
- uvicorn main:app --workers 4

What is Gunicorn?

- Python WSGI server
- Used for Flask & Django
- Manages worker processes
- No async support

Uvicorn vs Gunicorn

Uvicorn:

- ASGI
- Async support
- Best for FastAPI

Gunicorn:

- WSGI
- Sync
- Used for Flask/Django

Using Gunicorn with Uvicorn Workers

- In production:
- Gunicorn manages processes
- Uvicorn handles async requests

- Example:
- -gunicorn -k uvicorn.workers.UvicornWorker main:app

Accessing API from Browser

- http://127.0.0.1:8000
- http://localhost:8000

Common Errors

- Module not found
- app not defined
- Port already in use

Key Takeaways

- Uvicorn basics
- Server parameters
- Concept of Gunicorn
- Production awareness

What's Next?

- API Requests using Postman